



OTP Based Wireless Smart Lock System

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This is a one-time password (OTP) based smart locking system using Arduino and Bluetooth. It can generate a new password

every time you try to unlock it, which gives you extra security.

A key-based locking system has now become outdated because if your

key is lost or stolen, it creates a huge problem. Also, the electronic wireless locking system is not that safe because you might forget your password, or it might get stolen by others while entering the password on your device. So, we can make a DIY smart lock that has the capability to remove all these security threats and problems.

The author's prototype wired on a breadboard is shown in Fig. 1.

To start the project, we need the following hardware components/materials:

- Arduino Uno
- Bluetooth HC-05
- Jumper wires
- Servo motor
- LED
- 5V battery or power bank
- USB type-B cable for programming Arduino
- Breadboard or veroboard PCB

After collecting all these components, start the project by writing an Arduino code.

Coding

First, we need to include the library and declare variables, as shown in code snippet Fig. 2. Include a servo library and then create a string array to generate a password. Next, create a few more string variables to store the password, OTP, and LED pin numbers as given in code snippet.

Now, in the second code snippet,

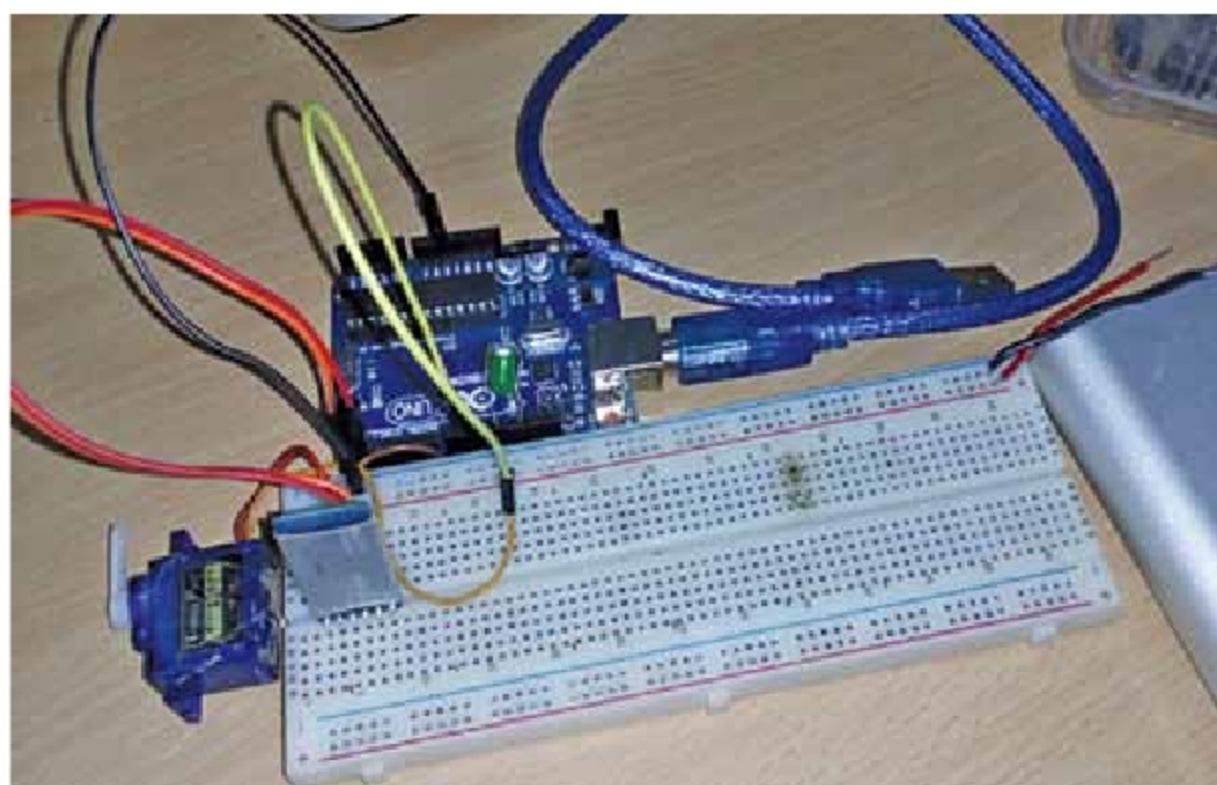


Fig. 1: Author's prototype wired on breadboard

```
#include <Servo.h>
String didid;
String pwd;
String letters[6] = {"adv", "fdfdb", "fdfc", "fdfd", "efdf", "fmbff"};
String otp = "";
String numbers[4]={"3213", "213213", "9999", "543646"};
int sled1=12;
int sled2=11;
Servo servo;
```

Fig. 2: Arduino code with library and variables

```
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
  servo.attach(9);
  pinMode(sled1,1);
  pinMode(sled2,1);
}
```

Fig. 3: Setup serial, servo and pin mode

```
void loop() {
  // put your main code here, to run repeatedly:
  while (Serial.available()==0);
  didid=Serial.readStringUntil('\n');
  if (didid=="asdfg"){
    otp();
    digitalWrite(sled1,1);
  }
  check();
}
```

Fig. 4: Loop() and OTP() functions